

8(a)	photon energy (to remove electron)	B1
	minimum energy to remove electron or energy to remove electron from surface or energy to remove electron with zero kinetic energy	B1
8(b)(i)	photon energy = hf	C1
	number per unit time = $8.36 \times 10^{-3} / (1.36 \times 10^{15} \times 6.63 \times 10^{-34})$ $= 9.27 \times 10^{15} \text{ s}^{-1}$	A1
8(b)(ii)	$hf = \phi + E_{\text{MAX}}$	C1
	$\phi = (1.36 \times 10^{15} \times 6.63 \times 10^{-34}) - (3.09 \times 10^{-19})$ $= 5.93 \times 10^{-19} \text{ J}$	A1
8(c)(i)	greater photon energy (and same work function)	M1
	so maximum kinetic energy is increased	A1
8(c)(ii)	(greater photon energy and same power so) lower number of photons (per unit time)	M1
	(each electron absorbs one photon) so lower rate of emission	A1